PART A Five chapters, in the order the module teaches them. Your reference for every video, lab, and exam question in Module 1.

Pre-work

1 min

PART B Four packets: Intro to Anatomy, Cell Anatomy, Tissues and Histology, Integumentary. Complete these by hand before the class day they belong to. These pages print landscape.

Lab Structure List

1 min

PART C The Module 1 slice of the master lab list. Everything here is testable on the lab portion of Exam 1.

Solano Community College · Fall 2026

DR. SHARILYN RENNIE

BIO 004 · MODULE 1

Contents

Five concepts, in the order the module teaches them. Part A of the Module 1 packet. Part B is the pre-work, Part C is the lab structure list.

Everything else for Module 1

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 1 COVERS

Module 1 runs across the first three weeks of the term and ends in Exam 1. It carries 23 competencies, listed under each section title by their Mastery OS identifiers.

The five concepts build on one another in order. The language of anatomy gives you the words. The body cavities give you the map. The cell is the unit every tissue is made from. Histology is the four tissues those cells build. The integumentary system is the first organ where all four appear at once, so it is where the module's ideas get used rather than listed.

This is anatomy. Where function appears in these notes it is marked as a physiology note, and it is there because the structure does not make sense without it.

Course Notes.

1 min

PART A

Five chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B. This is your reference for every video, lab, and exam question in Module 1.

SECTION 1

The Language of Anatomy

9 min

INTRODUCTION TO ANATOMY

The shared vocabulary the rest of the course is built on. Every position, plane, direction, and region named here gets used in lab, on exams, and in every clinical description you will ever read.

COMPETENCIES W1-LEVELS-ORGANIZATION, W1-ANATOMICAL-POSITION, W1-PLANES-SECTIONS, W1-DIRECTIONAL-TERMS, W1-REGIONAL-TERMS

BY THE END

1. Define anatomy and physiology and name the branches of each.
2. List the six levels of structural organization in order and give an example of each.
3. Name the eleven organ systems and the six basic life processes.
4. Place the body in anatomical position and explain why it is the reference for every directional term.
5. Name the planes and sections and use the paired directional terms precisely.
6. Identify the major anterior and posterior regional terms.
7. Separate disease from disorder and signs from symptoms.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Week 1

drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html

Anatomy and Physiology

Anatomy is the study of the structures in the body. **Physiology** is the study of how those structures function, how they work. The two are inseparable in practice. Structure sets the limits on what a part can do, and function explains why a structure is shaped the way it is. This course is the anatomy half.

Branches of anatomy

- **Gross (macroscopic) anatomy**, structures visible to the unaided eye. Studied **regionally** (all structures in one body area), **systemically** (one organ system at a time), or as **surface anatomy** (internal structures located by external landmarks).
- **Microscopic anatomy**, structures too small to see unaided. Includes **cytology** (cells) and **histology** (tissues).
- **Developmental anatomy**, structural change across the lifespan. **Embryology** is the subdivision covering development before birth.
- **Pathological anatomy**, structural change caused by disease.
- **Radiographic anatomy**, internal structure seen through imaging.

Branches of physiology

- **Neurophysiology**, function of the nervous system.
- **Endocrinology**, hormones and how they regulate function.
- **Cardiovascular physiology**, function of the heart and vessels.
- **Respiratory physiology**, function of the airways and lungs.
- **Renal physiology**, function of the kidneys.
- **Immunology**, how the body defends itself.
- **Pathophysiology**, functional change caused by disease.

Levels of Structural Organization

The body is organized from the simplest level to the most complex. Each level is built from the one before it, so a failure at a lower level always has consequences above it.

THE SIX LEVELS, LEAST TO MOST COMPLEX

Level	What it is	Example
Chemical	Atoms and molecules, the smallest level	Oxygen, water, a protein
Cellular	The smallest living unit of the body	A keratinocyte, a neuron
Tissue	Groups of similar cells performing a specific function	Stratified squamous epithelium
Organ	A structure composed of two or more tissue types with a defined job	The skin, the stomach
Organ system	Organs working together toward one broad function	The integumentary system
Organismal	All organ systems together, the entire living being	The person

The Eleven Organ Systems

SYSTEM, MAJOR COMPONENTS, PRIMARY FUNCTION

System	Major components	Primary function
Integumentary	Skin, hair, nails, cutaneous glands	Covers and protects, regulates temperature, senses the environment
Skeletal	Bones, cartilage, ligaments, joints	Supports, protects, stores minerals, houses blood formation
Muscular	Skeletal muscles and their tendons	Produces movement, maintains posture, generates heat
Nervous	Brain, spinal cord, nerves, sensory receptors	Fast control, detects and responds to change
Endocrine	Pituitary, thyroid, adrenals, pancreas, gonads	Slow control through hormones
Cardiovascular	Heart, blood vessels, blood	Transports oxygen, nutrients, wastes, and heat
Lymphatic and immune	Lymph vessels and nodes, spleen, thymus, tonsils	Returns fluid, filters lymph, mounts immune defense
Respiratory	Nasal cavity, pharynx, larynx, trachea, bronchi, lungs	Gas exchange between air and blood
Digestive	Alimentary canal plus liver, gallbladder, pancreas	Breaks down food, absorbs nutrients, eliminates residue
Urinary	Kidneys, ureters, bladder, urethra	Filters blood, eliminates nitrogenous waste, balances water and electrolytes
Reproductive	Gonads, ducts, accessory glands, external genitalia	Produces gametes and offspring

The Six Basic Life Processes

These distinguish living things from non-living things. Every one of them shows up again in a later unit.

1. **Metabolism**, the sum of all chemical reactions in the body. **Catabolism** breaks molecules down, **anabolism** builds them up.
2. **Responsiveness**, the ability to detect and react to a change in the internal or external environment.
3. **Movement**, motion of the whole body, of individual organs, of single cells, or of structures inside a cell.
4. **Growth**, an increase in size, from more cells, larger cells, or more material between cells.
5. **Differentiation**, the change of an unspecialized cell into a specialized one. Stem cells become keratinocytes, neurons, muscle fibers.
6. **Reproduction**, either the formation of new cells for growth and repair, or the production of a new individual.

Anatomical Position

Anatomical position is the standard reference posture: the body erect, feet flat and slightly apart, head and eyes facing forward, arms at the sides, palms facing forward.

Why it matters. Every directional term assumes this position. That way a description does not change with how the person is actually standing, sitting, or lying. Right and left are always the patient's right and left, not yours.

Two related terms describe a body lying down. **Supine** is lying face up, on the back. **Prone** is lying face down, on the front.

Body Planes and Sections

A **plane** is an imaginary flat surface passing through the body. The **section** is the cut surface that plane produces. The **midline** is the imaginary vertical line running from the top of the head, through the nose, down the spine, to the pelvis. It divides the body into equal left and right halves and serves as the reference point for describing locations relative to the body's center.

PLANES AND WHAT EACH ONE DIVIDES THE BODY INTO

Plane	Orientation	Divides the body into
Sagittal	Vertical	Left and right parts
Midsagittal (median)	Vertical, on the midline	Equal left and right halves
Parasagittal	Vertical, off the midline	Unequal left and right parts
Frontal (coronal)	Vertical	Anterior and posterior portions
Transverse (horizontal)	Horizontal	Superior and inferior portions, a cross section
Oblique	At an angle other than 90 degrees	Angled portions, between horizontal and vertical

Directional Terms

Directional terms come in pairs of opposites, and they are always given from the patient's perspective in anatomical position. A structure lying between a medial and a lateral structure is called **intermediate**.

PAIRED DIRECTIONAL TERMS

Term	Its opposite	The relationship the pair describes
Superior	Inferior	Toward the head or above, versus away from the head or below
Anterior (ventral)	Posterior (dorsal)	Toward the front, versus toward the back of the body
Medial	Lateral	Toward the midline, versus away from the midline
Proximal	Distal	Closer to the point of attachment to the trunk, versus farther from it. Limbs only
Superficial	Deep	Toward the body surface, versus away from the surface
Central	Peripheral	Near the center of a structure, versus away from its center
Ipsilateral	Contralateral	On the same side of the body, versus on the opposite side

Regional Terms

The body divides into an **axial** region, the head, neck, and trunk, and an **appendicular** region, the limbs. Each named region has a precise term. Learn these as a location on your own body, not as a word list.

Anterior view

ANTERIOR REGIONAL TERMS

Term	Region	Term	Region
Frontal	Forehead	**Inguinal**	Groin
Cranial	Skull or head	**Pubic**	Area near the genital region
Facial	Face	**Brachial**	Upper arm
Orbital or ocular	Eye region	**Antebrachial**	Forearm
Buccal	Cheek	**Antecubital**	Front of the elbow
Otic	Ear region	**Carpal**	Wrist
Nasal	Nose	**Pollex**	Thumb
Oral	Mouth region	**Palmar**	Palm of the hand
Mental	Chin	**Digital or phalangeal**	Fingers or toes
Cervical	Neck region	**Patellar**	Kneecap
Thoracic	Chest area	**Crural**	Leg, the shin area
Mammary	Breast region	**Tarsal**	Ankle
Abdominal	Below the chest and above the pelvis	**Pedal**	Foot
Umbilical	Navel or belly button	**Hallux**	Big toe
Pelvic	Lower abdomen or hip area		

Posterior view

POSTERIOR REGIONAL TERMS

Term	Region	Term	Region
Cephalic	Head	**Sacral**	Area between the hips
Cervical	Neck	**Gluteal**	Buttock region
Acromial	Point of the shoulder	**Femoral**	Thigh
Dorsal	Back	**Popliteal**	Back of the knee
Olecranal	Back of the elbow	**Sural**	Calf
Lumbar	Lower back or loin area	**Calcaneal**	Heel
Plantar	Sole of the foot		

Clinical Vocabulary

Disease versus disorder

Disease is a pathological condition caused by identifiable physical or biochemical changes, often from infection, genetics, or environmental factors. A disease typically has a clear set of symptoms, a predictable progression, and an identifiable cause. Examples: diabetes mellitus, a metabolic disease involving insulin production or utilization. Tuberculosis, an infectious disease caused by **Mycobacterium tuberculosis**.

Disorder is a disruption of normal physical or mental function that often lacks a single clear cause. Disorders may result from complex interactions among genetic, environmental, and lifestyle factors. Examples: anxiety disorder, marked by excessive fear or worry. Functional bowel disorders such as irritable bowel syndrome, which have no clear structural abnormality.

TWO MORE WAYS TO SORT A CONDITION

Term	Meaning	Example
Local	Affects one part or one limited region of the body	A skin abscess
Systemic	Affects the whole body or several organ systems at once	Sepsis, influenza
Acute	Sudden onset, short duration	Appendicitis
Chronic	Gradual onset, long duration, often lifelong	Type 2 diabetes mellitus

Signs versus symptoms

A **sign** is objective. It is a measurable physical manifestation that a healthcare professional can observe. Fever measured as an elevated temperature, a visible rash, an increased heart rate on a pulse check or ECG.

A **symptom** is subjective. It is the patient's own experience of illness, and it cannot be measured directly by an observer. Pain, fatigue, nausea.

THE SAME CONDITION, SORTED BOTH WAYS

Condition	Signs (observed)	Symptoms (reported)
Influenza	Fever, nasal discharge, coughing	Fatigue, body aches, sore throat
Anxiety disorder	Increased heart rate, sweating during episodes	Feeling of dread, difficulty concentrating

What goes into a clinical diagnosis

- **History**, what the patient reports. Symptoms, timing, what makes it better or worse, past conditions, family history.
- **Physical examination**, the signs the clinician can observe, palpate, percuss, or listen for.
- **Vital signs**, temperature, pulse, respiratory rate, blood pressure, oxygen saturation.
- **Laboratory testing**, blood, urine, culture, tissue.
- **Imaging**, radiograph, CT, MRI, ultrasound. The plane you image in determines what you can see.
- **Differential diagnosis**, the ranked list of conditions that could explain the findings, narrowed as evidence comes in.

WHY THE WORDS ARE NOT OPTIONAL

Anatomical terms remove ambiguity. A chart that reads "wound on the anterior, distal forearm" means the same thing to every clinician on every shift, no matter how the patient is positioned. Imaging is built on the planes: a CT slice is a transverse section, and an MRI is often read in the sagittal and frontal planes, so knowing the plane tells you which view you are looking at. Paired terms have to be used precisely. Proximal and distal describe the limbs only, never the trunk. Ipsilateral and contralateral matter in neurology, where a lesion on one side of the brain can show its effect on the opposite side of the body.

SECTION 2

Body Cavities & Regions

5 min

INTRODUCTION TO ANATOMY

The body cavities enclose and organize the internal organs. Knowing which cavity a structure sits in tells you what membrane wraps it, what fluid surrounds it, and what happens when that space is invaded.

COMPETENCIES W1-BODY-CAVITIES, W1-SEROUS-MEMBRANES, W1-MEDIASTINUM, W1-PERITONEAL-RELATIONSHIPS, W1-ABDOMINOPELVIC-REGIONS

BY THE END

1. Name the two major body cavities and their subdivisions, and state what each contains.
2. Define a serous membrane and name the three membrane pairs, their spaces, and their fluids.
3. Give the boundaries and contents of the mediastinum.
4. Define retroperitoneal and list the retroperitoneal structures using SAD PUCKER.
5. Draw the four quadrants and the nine regions and name the organs in each.
6. Connect an inflamed serous membrane to the condition it produces.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Week 1

drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html

The Two Major Body Cavities

A **body cavity** is a space within the body that houses and protects internal organs. There are two major cavities, and each divides further.

CAVITIES, SUBDIVISIONS, AND CONTENTS

Major cavity	Subdivision	Contents
Anterior (ventral)	Thoracic cavity	The lungs and the heart
Anterior (ventral)	Abdominopelvic cavity	The abdominal and pelvic organs
Posterior (dorsal)	Cranial cavity	The brain
Posterior (dorsal)	Spinal (vertebral) cavity	The spinal cord

The boundary. The **diaphragm** is the muscular partition that separates the thoracic cavity above from the abdominopelvic cavity below. It is the single most useful landmark in the ventral cavity, because it tells you which membranes and which fluids you are dealing with.

Serous Membranes

A **serous membrane** is a membrane that lines a cavity lacking an opening to the outside and covers the internal organs within it. Every serous membrane is a double layer folded back on itself.

- **Parietal layer**, lines the inner body wall.
- **Visceral layer**, covers the surface of an organ.
- **Serous space**, the cavity between the two layers.
- **Serous fluid**, found in that space between the visceral and parietal layers. It reduces friction so the organ can move against the wall without damage.

THE THREE NAMED SEROUS MEMBRANE PAIRS

Membrane	Wraps	Space	Fluid
Pericardium	The heart (parietal and visceral pericardium)	Pericardial cavity	Pericardial fluid
Pleura	The lungs (parietal and visceral pleura)	Pleural cavity	Pleural fluid
Peritoneum	The abdomen (parietal and visceral peritoneum)	Peritoneal cavity	Peritoneal fluid

HOLD ONTO THIS

Parietal goes with the wall. Visceral goes with the organ. The two words never change meaning, no matter which cavity you are in.

The Mediastinum

The **mediastinum** is the central compartment of the thoracic cavity, the region between the two lungs.

BOUNDARIES OF THE MEDIASTINUM

Boundary	Structure
Superior	Thoracic inlet
Inferior	Diaphragm
Anterior	Sternum
Posterior	Vertebral column
Lateral	Lungs

Contents: heart, trachea, esophagus, major blood vessels, thymus.

The Peritoneum and Retroperitoneal Structures

The **retroperitoneum** is the area posterior to the peritoneum but anterior to the spine. A structure that lies there is called **retroperitoneal**. A structure fully wrapped in visceral peritoneum and suspended within the peritoneal cavity is **intraperitoneal**. The distinction is surgical: it determines the approach a surgeon takes and how an infection or a bleed will spread.

MNEMONIC: SAD PUCKER

S Suprarenal (adrenal) glands. **A** Aorta and inferior vena cava. **D** Duodenum, second and third segments. **P** Pancreas, except the tail. **U** Ureters. **C** Colon, ascending and descending. **K** Kidneys. **E** Esophagus. **R** Rectum.

Abdominal Quadrants and Regions

The four quadrants

Two planes draw the quadrant grid: a median (midsagittal) plane and a transverse plane through the umbilicus. Quadrants are the fast clinical shorthand, the version used when someone is describing pain in a triage note.

THE FOUR QUADRANTS AND THEIR KEY ORGANS

Quadrant	Key organs
Right upper (RUQ)	Liver, gallbladder
Left upper (LUQ)	Stomach, spleen
Right lower (RLQ)	Appendix, cecum
Left lower (LLQ)	Descending colon

The nine regions

Four planes, two vertical and two horizontal, divide the abdominopelvic cavity into nine regions. Regions are the more precise anatomical version, the one used when the location has to be described exactly.

THE NINE ABDOMINOPELVIC REGIONS

Region	Key structure
Right hypochondriac	Liver, gallbladder
Epigastric	Stomach
Left hypochondriac	Spleen
Right lumbar	Ascending colon
Umbilical	Small intestine
Left lumbar	Descending colon
Right iliac	Cecum, appendix
Hypogastric	Urinary bladder
Left iliac	Sigmoid colon

Inflammatory Disorders of the Serous Membranes

Each named membrane has a matching inflammatory condition. If you know the membrane and its location, you can predict the presentation.

FOUR CONDITIONS, BY MEMBRANE

Condition	Causes	Signs and symptoms	Treatment
Peritonitis	Infection, trauma, ruptured appendix	Abdominal pain, fever, rigidity	Antibiotics, surgery
Pleurisy	Viral infection, pneumonia	Sharp chest pain, dry cough	Anti-inflammatory drugs
Pericarditis	Viral infection, autoimmune disorders	Chest pain, pericardial friction rub	NSAIDs, corticosteroids
Meningitis	Bacterial or viral infection	Stiff neck, fever, headache	Antibiotics for bacterial, supportive care for viral

THE CAVITY TELLS YOU THE CONSEQUENCES

A serous cavity is a closed space, so anything that enters it has nowhere to drain. A ruptured appendix spills bowel contents into the peritoneal cavity, and the result is peritonitis across the whole abdomen rather than a local problem. A puncture of the pleural cavity lets air in, the negative pressure that holds the lung expanded is lost, and the lung collapses. Reduced serous fluid in any of the three spaces means the two layers rub instead of glide, which is why a friction rub is audible in pericarditis. Meningitis is the same logic applied to the dorsal cavity: the meninges enclose the brain and spinal cord, so an infection there raises pressure inside a rigid skull.

SECTION 3

Anatomy of the Cell

6 min

CELL ANATOMY

The cell is the smallest living unit of the body. Every organelle in it is a structure with a job, and every one of those jobs has a disease that shows what happens when it fails.

COMPETENCIES W1-GENERALIZED-CELL, W1-PLASMA-MEMBRANE, W1-NUCLEUS, W1-ORGANELLE-ID

BY THE END

1. Identify the three major components of a cell and describe what each does.
2. Define semipermeable, permeable, and impermeable, with examples.
3. Name the structures of the nucleus and state the function of each.
4. Separate cytoplasm from cytosol.
5. Describe the plasma membrane and the fluid mosaic model.
6. List the components of the endomembrane system and trace a protein through it.
7. Describe each organelle and connect it to a clinical consequence when it fails.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The Cell

drsrennie-stack.github.io/new-build-bio4-solano/cell-concept-videos.html

General Structure of the Cell

Plasma membrane

A flexible, semipermeable boundary that encloses the cell, separates its contents from the external environment, and regulates the passage of substances. It is composed primarily of lipids and proteins with some carbohydrates, and it carries specialized structures such as receptors, ion channels, and transport proteins for communication and material exchange.

Nucleus

The largest organelle and the command center of the cell. It contains the genetic material, DNA, organized into chromosomes. It controls gene expression, cellular growth, and replication, and it is the site of transcription, the conversion of DNA to RNA.

Cytoplasm

The gel-like substance within the plasma membrane, excluding the nucleus, containing the organelles and the cytosol. It facilitates intracellular processes such as glycolysis and signal transduction, helps maintain cell shape, and enables the movement of materials.

Cell Permeability

THREE DEGREES OF PERMEABILITY

Term	Definition	Examples
Semipermeable	Allows selective substances to cross based on size, charge, and solubility. Facilitates passive diffusion for some molecules and relies on transport proteins for others	Oxygen, carbon dioxide, and small lipid-soluble molecules diffuse freely
Permeable	Permits the free passage of substances without restriction	Water through aquaporins, gases such as nitrogen
Impermeable	Blocks movement unless a specific mechanism carries the substance across	Proteins, polysaccharides, charged ions with no transporter

The Nucleus in Detail

Multinucleated cells have more than one nucleus. Skeletal muscle fibers and hepatocytes in the liver are the standard examples. **Anucleated cells** have none. Mature red blood cells are the example, and losing the nucleus is what makes room for hemoglobin.

Function. The nucleus stores the DNA that carries instructions for protein synthesis and cellular activities, plays a central role in cell division, growth, and metabolism, and regulates which genes are turned on or off in different cells. That last one is what allows differentiation.

STRUCTURES OF THE NUCLEUS

Structure	What it is and does
Nuclear envelope	A double-layered membrane separating the nucleus from the cytoplasm and maintaining a controlled environment
Nuclear pores	Protein-lined channels allowing exchange of materials such as RNA and proteins between nucleus and cytoplasm
Chromatin	A network of DNA and histone proteins that condenses into visible chromosomes during cell division
Chromosomes	Rod-shaped structures containing genes, responsible for passing genetic information during reproduction
Nucleoplasm	The viscous fluid within the nucleus supporting the chromatin and nucleolus
Nucleolus	A dense region where ribosomal RNA is synthesized and ribosomal subunits are assembled

Cytoplasm versus Cytosol

Cytoplasm is the entire contents of the cell excluding the nucleus, meaning the organelles suspended in cytosol. **Cytosol** is the aqueous component alone: water, ions, enzymes, and small organic molecules. The cytosol is where many metabolic reactions happen, glycolysis among them.

The Plasma Membrane in Detail

Function. A dynamic barrier that regulates interactions between the cell and its environment.

MEMBRANE COMPONENTS

Component	Arrangement and role
Phospholipid bilayer	Hydrophilic heads face outward toward the extracellular and intracellular fluids. Hydrophobic tails face inward, away from water
Proteins	Integral proteins span the bilayer, peripheral proteins bind to the surface. Roles: transport, enzymatic activity, signal transduction
Glycoproteins and glycolipids	Cell recognition and immune responses
Cholesterol	Provides stability and moderates membrane fluidity at varying temperatures

Fluid mosaic model. The membrane behaves as a fluid-like sea of phospholipids with proteins and carbohydrates floating within it. Proteins and lipids move laterally, which allows flexibility and dynamic interaction rather than a fixed wall.

The Endomembrane System

A coordinated network of organelles responsible for protein and lipid synthesis, modification, and transport.

- **Nuclear envelope**, protects the genetic material.
- **Rough ER**, synthesizes membrane-bound and secretory proteins.
- **Smooth ER**, synthesizes lipids, detoxifies chemicals, stores calcium.
- **Golgi apparatus**, modifies, sorts, and packages proteins and lipids.
- **Lysosomes**, digestive organelles containing hydrolytic enzymes.
- **Vesicles**, transport materials between organelles or to the plasma membrane.

Ribosomes

Function. The site of protein synthesis, where mRNA is translated into polypeptides. **Structure.** A large and a small subunit, each made of ribosomal RNA and protein.

- **Free ribosomes**, produce proteins for use inside the cell.
- **Bound ribosomes**, attached to the rough ER, producing proteins for secretion or for insertion into membrane.
- **Mitochondrial ribosomes**, produce proteins specific to mitochondrial function.

Endoplasmic Reticulum

ROUGH AND SMOOTH ER COMPARED

	Rough ER	Smooth ER
Structure	Studded with ribosomes	Lacks ribosomes
Function	Produces membrane proteins and secretory proteins	Synthesizes lipids, metabolizes carbohydrates, detoxifies drugs, stores calcium ions
Next stop	Sends vesicles to the Golgi apparatus	Supplies lipid and calcium to the rest of the cell

Golgi Apparatus

Function. Modifies, packages, and sorts proteins and lipids, tagging molecules for delivery to specific cellular locations. **Structure.** A series of flattened sacs called cisternae, with vesicles budding off.

Lysosomes

Function. Digestive organelles that break down macromolecules, damaged organelles, and pathogens. Two processes to know: **phagocytosis**, engulfing pathogens or debris, and **autophagy**, recycling of the cell's own components.

CLINICAL CORRELATION

In Tay-Sachs disease, a deficiency of a lysosomal enzyme causes toxic accumulation of lipids in nerve cells. The organelle that should be clearing material cannot, so the material builds up inside the cell until the cell fails.

Peroxisomes

Function. Break down fatty acids, amino acids, and hydrogen peroxide. They also detoxify alcohol in the liver and kidney, which is why those organs are peroxisome-rich.

Mitochondria

Function. The powerhouse of the cell, producing ATP through cellular respiration. **Structure.** A double membrane with inner folds called cristae, which multiply the surface area available for ATP production. Mitochondria contain their own DNA and ribosomes, so they produce some of their own proteins.

Cytoskeleton

Function. Provides structural support, facilitates intracellular transport, and enables cellular movement.

THREE CYTOSKELETAL COMPONENTS

Component	Made of	Role
Microfilaments	Actin	Cell shape and movement
Intermediate filaments	Various fibrous proteins	Mechanical support, resisting tension
Microtubules	Tubulin	Tubular structures forming centrioles, cilia, and flagella

Centrioles and Centrosomes

Centrioles are short cylindrical structures organized in a 9 plus 0 microtubule arrangement. The **centrosome** is the microtubule-organizing center. Both are essential for forming the mitotic spindle during cell division.

Cilia and Flagella

Cilia are short projections that move substances across the cell surface, such as mucus in the respiratory tract. **Flagella** are long projections that propel the cell itself, as in sperm. Both use a 9 plus 2 microtubule arrangement, and that arrangement is what makes movement possible.

HOLD ONTO THIS

9 plus 0 does not move, it organizes. 9 plus 2 moves. Centrioles are 9 plus 0. Cilia and flagella are 9 plus 2.

THE ORGANELLE POINTS TO THE DISEASE

Read this list backward when you are given a case. A cell that cannot break down lipids has a lysosome problem. A cell that cannot make enough ATP has a mitochondrial problem, and the tissues that fail first are the ones with the highest energy demand: muscle, brain, kidney. A cell that cannot clear mucus has a ciliary problem, and the presentation is recurrent respiratory infection. Structure first, then the consequence.

SECTION 4

Histology: The Four Tissue Types

10 min

TISSUES & HISTOLOGY

Tissues are groups of similar cells performing a specific function. Four types build every organ in the body, and the arrangement of the cells is what predicts what the tissue can do.

COMPETENCIES W1-GERM-LAYERS, W1-EPITHELIAL-ID, W1-CELL-JUNCTIONS, W1-CONNECTIVE-ID, W1-BODY-MEMBRANES

BY THE END

1. Name the four tissue types, their general functions, and where each is found.
2. Define a biopsy and walk through the procedure.
3. Identify the five cell junctions, their proteins, locations, and functions.
4. Separate epithelial from connective tissue on structure alone.
5. Classify epithelium by layers and cell shape, and name a location for each type.
6. Classify glands structurally and functionally.
7. Name the cells, fibers, and ground substance of connective tissue and what each contributes.
8. Recognize cartilage, bone, muscle, and nervous tissue at survey level.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Tissues

drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html

The Four Major Tissue Types

1. **Epithelial tissue**, covers body surfaces, lines cavities, and forms glands.
2. **Connective tissue**, protects, supports, and binds structures and organs.
3. **Nervous tissue**, detects changes and generates nerve impulses.
4. **Muscle tissue**, facilitates movement and generates force.

Cells that stay and cells that move

Most cells are **stationary**. Some **migrate**, particularly during growth and development. White blood cells migrating to a site of injury or infection are the everyday example.

Biopsy

Definition. Removal of a tissue sample for microscopic examination. This is how tissue-level questions get answered clinically.

1. Collect a tissue sample using a needle, scalpel, or endoscope.
2. Preserve the tissue in a fixative such as formalin.
3. Section the tissue into thin slices for examination.
4. Stain the tissue for contrast under a microscope.

The Five Cell Junctions

JUNCTION, PROTEIN, LOCATION, FUNCTION

Junction	What it is	Function	Location	Major protein
Tight junction	A network of transmembrane proteins that fuse together on adjacent plasma membranes, sealing the passageway between cells	Prevents substances passing between cells, prevents organ contents leaking into surrounding tissue	Epithelium lining the stomach, intestines, urinary bladder, blood-brain barrier	Claudins and occludins
Adherens junction	Adhesion belts formed by transmembrane proteins attached to microfilaments	Provides mechanical stability, helps epithelial surfaces resist separation during contractile activity	Intestinal epithelial cells	Cadherins
Desmosome	Plaques and cadherins connected to intermediate filaments	Strongly binds cells to each other, prevents separation under tension	Outer epidermis, cardiac muscle cells	Cadherins
Hemidesmosome	Anchors cells to the basement membrane	Prevents separation of epidermis from basement membrane, stabilizes epithelial cells during mechanical stress	Skin, heart muscle	Integrins
Gap junction	Protein channels connecting neighboring cells, allowing passage of small ions and molecules	Cell to cell communication, transfers nutrients and waste, allows impulses to spread	Lens, cornea, embryonic tissues, neurons, cardiac tissue, GI tract, uterus	Connexins

HOLD ONTO THIS

Desmosomes connect cell to cell. Hemidesmosomes connect cell to basement membrane. Half the name, half the job.

Epithelium versus Connective Tissue

THE TWO TISSUE TYPES COMPARED

Feature	Epithelial tissue	Connective tissue
Matrix and cells	Tightly packed cells, minimal extracellular matrix	Large extracellular matrix, few cells
Blood vessels	Avascular, relies on diffusion	Highly vascular, with cartilage and tendon as exceptions
Location	Covers surfaces and forms linings	Supports and connects other tissues

Similarities. They work together for nutrient exchange, since avascular epithelium depends on the vascular connective tissue beneath it. Both play critical roles in maintaining structure and function.

Epithelial Tissue

Three functions of epithelium

- **Selective barrier**, controls transfer of substances into and out of the body.
- **Secretory surface**, produces and releases cellular products.
- **Protective surface**, provides resistance against abrasion and environmental influences.

The basement membrane

Location. Below the basal surface of epithelial tissue. **Composition.** A **basal lamina** of collagen, laminin, and glycoproteins, plus a **reticular lamina** of collagen and glycoproteins.

Functions. Supports and anchors epithelial tissue, facilitates wound healing and cell migration, restricts large molecules, and filters substances in the kidneys.

Classification of epithelium

EPITHELIAL TYPES, FUNCTION, AND LOCATION

Tissue type	Description	Function	Locations
Simple squamous	Single layer of flat, thin cells	Filtration, diffusion, osmosis, secretion	Alveoli, glomeruli, lining of blood vessels, serosa
Simple cuboidal	Single layer of cube-shaped cells	Secretion and absorption	Kidney tubules, small glands, ovary surface
Simple columnar	Single layer of tall, column-shaped cells, may have microvilli or goblet cells	Absorption, secretion of mucus and enzymes	Digestive tract from stomach to rectum, gallbladder
Stratified squamous	Several layers, apical layers are flat	Protection against abrasion	Skin (keratinized), esophagus, mouth, vagina
Stratified cuboidal	Two or more layers of cube-shaped cells	Protection and limited secretion	Sweat glands, mammary glands, salivary glands
Stratified columnar	Several layers, apical cells are columnar	Protection and secretion	Male urethra, ducts of some glands
Transitional	Several layers, cells stretch and change shape from dome-shaped to flat	Stretching to allow distension	Urinary bladder, ureters, part of the urethra
Pseudostratified ciliated columnar	Appears stratified but all cells touch the basement membrane, may have cilia and goblet cells	Secretion of mucus, propulsion by ciliary action	Respiratory tract (trachea, bronchi), sperm ducts

HOLD ONTO THIS

Name the layers first, then the shape of the apical cells. Simple means one layer, stratified means more than one, and the shape you name is always the shape at the free surface.

Glandular Epithelium

Structural classification

GLANDS BY NUMBER OF CELLS

Type	Description	Example
Unicellular	Single-celled glands	Goblet cells, which secrete mucus
Multicellular	Multiple cells forming a structure	Salivary glands, sweat glands

Duct type. Simple means no branching. Compound means branching. **Shape of the secretory portion.** Tubular means tube-shaped. Acinar means rounded. Combine the two and you have the full structural name.

Endocrine and exocrine

Endocrine glands release their product into the bloodstream, with no duct. Thyroid, adrenal, pituitary. **Exocrine glands** secrete their products into ducts that empty onto a surface or into a lumen. Sweat, salivary, sebaceous. Some glands are **mixed**, doing both, and the pancreas is the classic example: exocrine digestive enzymes into a duct, endocrine insulin and glucagon into the blood.

Functional classification of exocrine glands

HOW THE PRODUCT ACTUALLY LEAVES THE CELL

Type	Mechanism	Example
Merocrine	Secretions released via exocytosis. The cell is unharmed. Most common type	Salivary glands, pancreas
Apocrine	Secretions accumulate at the apical surface, which pinches off to release the product. The cell then repairs itself	Mammary glands
Holocrine	Secretions accumulate in the cytoplasm and the entire cell ruptures to release the product	Sebaceous glands

MEMORY TOOL

HOL equals WHOLE cell is secreted.

Connective Tissue

Functions

- Binds together, supports, and strengthens body tissues.
- Protects and insulates internal organs.
- Compartmentalizes structures, as in skeletal muscle.
- Serves as a major transport system. Blood is a connective tissue.
- Is the primary location of stored energy reserves, as in adipose tissue.
- Is a main source of immune responses.

The two basic elements

Connective tissue is **extracellular matrix** plus **cells**. The matrix divides into **ground substance** and **protein fibers**. The proportions of these elements are what make blood liquid, cartilage firm, and bone hard.

Ground substance

Definition. The material between cells and protein fibers. **Structure.** Fluid, semifluid, gelatinous, or calcified. **Composition.** Water and organic molecules such as chondroitin sulfate and glucosamine.

Functions. Supports and binds cells, supplies water and nutrients for exchange between blood and cells, and plays a role in tissue development and metabolism.

The matrix determines the quality. Cartilage is firm but pliable. Bone is hard and inflexible. Same basic plan, different matrix.

The three fiber types

COLLAGEN, RETICULAR, AND ELASTIC FIBERS

Fiber	Protein	Properties	Locations and functions
Collagen	Collagen, the most abundant protein in the body at about 25 percent	Very strong, resists tension. Parallel bundles provide tensile strength	Most connective tissues: bone, tendon, cartilage
Reticular	Collagen covered with glycoproteins	Fine, branching networks	Walls of blood vessels, surrounding cells in areolar, adipose, nervous, and smooth muscle tissue. Provides support and strength, forms the stroma of soft organs, helps form the basement membrane
Elastic	Elastin, with the glycoprotein fibrillin providing tensile strength and stability	Stretches up to 150 percent of relaxed length and returns to original shape	Skin, lungs, blood vessel walls

Cells of connective tissue

WHO LIVES IN THE MATRIX

Cell	Origin or definition	Function and location
Fibroblasts	Found in all general connective tissues	Migrate through connective tissue secreting matrix fibers and ground substance
Macrophages	Irregular white blood cells derived from monocytes	Engulf bacteria and debris via phagocytosis. Fixed macrophages stay in one tissue such as liver, lungs, spleen. Wandering macrophages migrate to sites of infection or inflammation
Mast cells	Found along blood vessels supplying connective tissue	Secrete histamine, which dilates blood vessels during inflammation. Also bind, ingest, and kill bacteria
Plasma cells	Develop from B lymphocytes	Secrete antibodies. Found in connective tissue of the GI tract, respiratory tract, and lymphatic tissues
Adipocytes	Connective tissue cells that store triglycerides	Found deep in the skin and around organs such as the heart and kidneys
White blood cells	Various leukocytes	Immune defense

Blasts versus cytes

IMMATURE AND MATURE FORMS

	Immature, the blasts	Mature, the cytes
Examples	Fibroblasts, osteoblasts, chondroblasts	Fibrocytes, osteocytes, chondrocytes
What they do	Actively divide and secrete extracellular matrix	Maintain and repair extracellular matrix

General features of connective tissue

- **Location.** Found below epithelial tissue, never on body surfaces.
- **Vascularity.** Highly vascular. Exceptions: cartilage and tendon.
- **Innervation.** Supplied with nerves. Exceptions: cartilage and blood.

Embryonic mesenchyme and mucous connective tissue

THE TWO EMBRYONIC CONNECTIVE TISSUES

	Embryonic mesenchyme	Mucous connective tissue
Location	In the embryo, under skin and along bones	In the umbilical cord
Significance	Differentiates into all types of connective tissue	Provides support
Characteristics	Irregular cells in a semifluid matrix with delicate reticular fibers	Widely scattered fibroblasts in a jelly-like matrix

Blood, a Fluid Connective Tissue

FORMED ELEMENTS AND WHAT EACH DOES

Cell type	Function
Neutrophils	Phagocytosis of bacteria
Eosinophils	Combat parasitic infections and allergies
Lymphocytes	Fight viruses
Monocytes and macrophages	Phagocytosis, transform into macrophages
Basophils	Involved in allergic responses
Red blood cells	Transport oxygen and carbon dioxide
Platelets	Facilitate blood clotting

Connective Tissues, Cartilage, Bone, and Nervous Tissue

SURVEY OF THE REMAINING TISSUE TYPES

Tissue type	Description	Function	Locations
Areolar connective	Loose arrangement of collagen, elastic, and reticular fibers with scattered cells	Strength, elasticity, support	Under epithelial tissue, around organs and blood vessels
Adipose connective	Cells filled with fat droplets, sparse extracellular matrix	Energy storage, insulation, cushioning	Subcutaneous tissue, around kidneys and heart
Hyaline cartilage	Glassy appearance, fine collagen fibers not visible with ordinary staining, chondrocytes in lacunae	Smooth surface for joint movement, flexible support	Ends of long bones, costal cartilage, nose, trachea, larynx
Fibrocartilage	Thick collagen fibers dominate	Resists compression, absorbs shock	Intervertebral discs, pubic symphysis, knee menisci
Elastic cartilage	Similar to hyaline but with more elastic fibers	Maintains shape and provides flexibility	External ear (pinna), epiglottis
Compact bone	Hard, calcified matrix with collagen fibers, osteocytes in lacunae	Supports, protects, stores calcium, houses blood formation	Bones
Nervous tissue	Neurons and glial cells, branching neurons with long processes	Transmission of electrical signals	Brain, spinal cord, nerves

Muscle Tissue

THE THREE MUSCLE TISSUE TYPES

Tissue type	Description	Function	Locations
Skeletal muscle	Long, cylindrical, multinucleated cells, striated	Voluntary movement, heat production	Attached to bones and occasionally skin
Cardiac muscle	Branching cells connected by intercalated discs, striated, uninucleated	Involuntary contraction to pump blood	Heart walls
Smooth muscle	Spindle-shaped cells, non-striated, uninucleated	Involuntary movement, propels substances	Walls of hollow organs such as intestines and bladder

TISSUE STRUCTURE PREDICTS HOW IT HEALS AND HOW IT FAILS

Vascularity sets healing speed. Cartilage and tendon are avascular or nearly so, which is why a torn meniscus or a ruptured Achilles heals slowly while a skin wound closes in days. Junction type sets what tears. Loss of desmosomal adhesion in the epidermis produces blistering, because the cells separate from each other. Loss of hemidesmosomes separates the whole epidermis from the basement membrane, which is a deeper and more serious blister. Cell arrangement sets the barrier. Stratified squamous epithelium survives abrasion because it sheds its top layer. Simple squamous cannot, which is why the alveoli, one cell thick for gas exchange, are so easily damaged.

SECTION 5

The Integumentary System

14 min

INTEGUMENTARY & SKIN

The skin is the largest organ in the body and the first place all four tissue types show up working together. It is also the organ you can read from across the room, which makes it the best teacher of the structure to consequence habit.

COMPETENCIES W1-SKIN-LAYERS, W1-EPIDERMAL-STRATA, W1-DERMIS-LAYERS, W1-SKIN-ACCESSORY

BY THE END

1. Name the components and the six functions of the integumentary system.
2. List the strata of the epidermis deep to superficial and say which is missing in thin skin.
3. Name the four principal cells of the epidermis and what each does.
4. Describe melanin transfer and why it protects the nucleus.
5. Separate the papillary from the reticular region of the dermis.
6. Describe hair anatomy, the follicle, and the bulb.
7. Compare the four skin glands on distribution, secretion, and onset.
8. Name the components of a nail and their functions.
9. Read skin color as a diagnostic clue.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Integumentary

drsrennie-stack.github.io/new-build-bio4-solano/integumentary-concept-videos.html

Components and Functions

Components. The skin, also called the cutaneous membrane, plus associated structures: hair, oil glands, sweat glands, sensory receptors, and nails.

Dermatology is the medical specialty dealing with the structure, function, and disorders of the integumentary system.

THE SIX FUNCTIONS OF THE SKIN

Function	How the structure delivers it
Thermoregulation	Sweat glands and dermal blood vessels adjust heat loss
Blood reservoir	Extensive dermal vascular network holds a substantial volume of blood
Protection	Keratin, lamellar granules, melanin, and immune cells
Cutaneous sensations	Tactile discs, corpuscles of touch, hair root plexuses, free nerve endings
Excretion and absorption	Small amounts of waste out, lipid-soluble substances in
Synthesis of vitamin D	UV exposure begins the synthesis pathway in the skin

General information. The skin is the largest organ by weight, covering the external surface. In an adult the area is roughly 22 square feet, and it weighs 10 to 11 pounds, about 7 percent of total body weight.

The Three Layers

1. **Epidermis**, part of the skin.
2. **Dermis**, part of the skin.
3. **Subcutaneous tissue**, not part of the skin.

The Epidermis

The thinner, superficial portion, composed of epithelial tissue and avascular. Its composition is **keratinized stratified squamous epithelium**. Avascular matters: the epidermis depends entirely on diffusion from the dermis beneath it, which is why a deep burn that destroys dermal vessels cannot support regeneration.

The four principal cells

CELLS OF THE EPIDERMIS

Cell	Share and origin	Function
Keratinocytes	90 percent, arranged in 4 to 5 layers	Produce keratin, a tough fibrous protein protecting skin and underlying tissue from abrasion, heat, microbes, and chemicals. Also produce lamellar granules, which release a water-repellent sealant that decreases water entry and loss and inhibits entry of foreign material
Melanocytes	8 percent, develop from the ectoderm of the embryo	Produce melanin, a yellow-red or brown-black pigment contributing to skin color and absorbing damaging UV light
Intraepidermal macrophages (Langerhans cells)	Originate in red bone marrow	Migrate to the epidermis and participate in immune responses against microbes invading the skin. Help other immune cells recognize and destroy an invader. Easily damaged by UV light
Merkel cells (tactile cells)	Least numerous, located in the deepest layer of the epidermis	Contact the flattened process of a sensory neuron, called a Merkel disc or tactile disc

The strata

Most regions of the body have **four strata**: stratum basale, stratum spinosum, stratum granulosum, stratum corneum. Areas of greatest friction, meaning fingertips, palms, and soles, have **five strata**, adding the stratum lucidum between the granulosum and the corneum.

MNEMONIC

Baby Screams Grandpa Loves Coffee. Basale, Spinosum, Granulosum, Lucidum, Corneum. Deep to superficial.

Melanin transfer

1. Melanocytes have long slender projections that extend between keratinocytes and transfer melanin granules to them.
2. Once inside the keratinocyte, melanin granules cluster to form a protective veil over the nucleus, on the side facing the skin surface.
3. That veil shields the DNA, preventing UV damage from the sun.
4. Melanocytes themselves remain susceptible to damage by UV light. They protect everyone else and not themselves.

The Dermis

The second, deeper portion of the skin. **Tissue composition.** Dense irregular connective tissue with collagen and elastic fibers.

Functions. Gives the skin great tensile strength, and allows stretch and recoil.

Characteristics. Thicker than the epidermis and variable by body site, greatest at the soles and palms. Contains blood vessels, nerves, glands, and hair follicles. **Cells.** Fibroblasts, a few macrophages, and a few adipocytes near the boundary with the subcutaneous layer. The dermis is essential to the survival of the epidermis and its associated structures.

THE TWO PRINCIPAL LAYERS OF THE DERMIS

	Papillary region	Reticular region
Position and size	Thinner, superficial. One fifth of the total dermal layer	Thicker, deep. Four fifths of the total dermal layer
Contents	Thin collagen and fine elastic fibers. Dermal papillae, small finger-shaped projections into the undersurface of the epidermis	Bundles of thick collagen fibers, scattered fibroblasts, wandering macrophages, adipocytes in the deepest part with coarse elastic fibers
Structures within	Capillary loops, Meissner corpuscles for touch and pressure, free nerve endings for temperature, light touch, and itching	Blood vessels, nerves, hair follicles, sudoriferous sweat glands, sebaceous glands
Function	Anchors epidermis, supplies it by diffusion, carries the fine touch receptors	Net-like layer attached to the subcutaneous layer. Provides strength, extensibility, and elasticity. The regular net-like collagen orientation makes skin resistant to stretching

Epidermal ridges, the fingerprints

The bond between dermis and epidermis on the surfaces of the palms, soles, and fingertips. **Functions.** Increase surface area, which increases the number of corpuscles of touch and therefore tactile sensitivity, and increase grip by increasing friction.

The Subcutaneous Layer (Hypodermis)

Deep to the dermis and not part of the skin. **Composition.** Areolar and adipose tissue. Fibers extend from the dermis and anchor the skin to this layer and to the fascia around muscles and bones. **Function.** Storage for fat. **Associated components.** Lamellated corpuscles, which are nerve endings sensitive to pressure, and large blood vessels.

Skin Color

THREE PIGMENTS

Pigment	Color contributed	Notes
Melanin	Pale yellow to reddish-brown	Two forms: pheomelanin, yellow to red, and eumelanin, brown to black
Hemoglobin	Pink to red	Seen through the epidermis, depends on the oxygen content of blood in the dermal capillaries
Carotene	Yellow-orange	A precursor to vitamin A, used to synthesize pigments for vision

Melanocyte distribution. Most numerous in the epidermis of darker areas such as the nipples and areolae, and in the face, limbs, and mucous membranes. The number of melanocytes is about the same in all people. The difference in skin color comes from the amount of pigment those cells produce and transfer to keratinocytes.

Melanin synthesis. Made from the amino acid tyrosine in the presence of the enzyme tyrosinase, inside an organelle called the **melanosome**. Exposure to UV light increases the activity of that enzyme, which increases melanin production. That is a tan, and it is protective against further exposure.

Hair

Location. Most skin surfaces. Exceptions: palms, fingertips, soles, and plantar surfaces of the feet. **Determinants of thickness and distribution.** Genetic and hormonal.

Functions. Limited protection from the sun, decreased heat loss from the scalp, eyelashes and eyebrows protecting the eyes from debris, nostril and ear hair protecting those structures from debris, and sensing light touch via hair root plexuses.

Anatomy of a hair

Composition. Dead, keratinized epidermal cells. **Two parts.** The **hair root**, the portion penetrating into the dermis and sometimes the subcutaneous tissue, and the **hair shaft**, the portion above the surface of the skin. Both have the same three concentric layers.

1. **Medulla**, the inner layer, 2 to 3 rows of irregular cells containing pigment.
2. **Cortex**, the major part of the shaft, elongated cells.
3. **Cuticle**, the outermost, a single layer of thin, flat, heavily keratinized cells.

The hair follicle

The follicle surrounds the root of the hair.

- **External root sheath**, a downward continuation of the epidermis.
- **Internal root sheath**, produced by the matrix, forming a cellular tubular sheath of epithelium. External plus internal root sheath equals the **epithelial root sheath**.
- **Dermal root sheath**, the dermis surrounding the hair follicle.

The bulb is the onion-shaped structure at the base of each follicle, surrounding the dermal root sheath. It contains the **papilla of the hair**, which is areolar connective tissue with many blood vessels, and the **hair matrix**, the germinal layer of cells arising from the stratum basale. The matrix is the site of cell division: it is responsible for the growth of existing hairs, the generation of new hairs when old ones are shed within the same follicle, and it gives rise to the cells of the internal root sheath.

Structures associated with hair

- **Sebaceous oil glands**, secrete oil onto the hair.
- **Arrector pili muscles**, smooth muscle that pulls the hair shaft perpendicular, producing goose bumps.
- **Hair root plexus**, dendrites of neurons surrounding each follicle, detecting light touch.

Types of hair and hair color

Lanugo is the very fine, non-pigmented downy hair covering the fetus. **Terminal hairs** are long, coarse, heavily pigmented hairs that replace lanugo just before birth on the eyebrows, eyelashes, and scalp. **Vellus hairs**, the peach fuzz, replace lanugo on the rest of the body. After puberty, under hormonal influence, some vellus hairs are replaced by terminal hairs, and the proportions differ between individuals.

Hair color is determined by the amount and type of melanin in the keratinized cells. Melanin is synthesized by melanocytes, scattered in the matrix of the bulb, and passes into the cells of the cortex and medulla. Dark hair is mostly eumelanin. Blond and red hair are mostly pheomelanin. Gray hair reflects a progressive decline in melanin production and contains few melanin granules. White hair lacks melanin entirely and accumulates air bubbles in the shaft.

Skin Glands

THE FOUR GLANDS COMPARED

Gland	Distribution	Secretory portion	Duct terminates	Secretion	Onset
Sebaceous (oil)	Largely lips, glans penis, labia minora, tarsal glands. Small on trunk and limbs. Absent on palms and soles	Dermis	Mostly connected to a hair follicle	Sebum, a mixture of triglycerides, cholesterol, proteins, and inorganic salts	Relatively inactive through childhood, activated at puberty
Eccrine sweat	Throughout most regions of the body, especially forehead, palms, and soles	Mostly deep dermis, sometimes the upper subcutaneous layer	Surface of the epidermis	Perspiration: water, ions (sodium, chloride), urea, uric acid, ammonia, amino acids, glucose, lactic acid	Soon after birth
Apocrine sweat	Skin of axillae, groin, areolae, bearded regions of the face, clitoris and labia minora	Mostly deep dermis and upper subcutaneous layer	Hair follicles	Perspiration with the same components as eccrine sweat plus lipids and proteins	Puberty
Ceruminous	External auditory canal	Subcutaneous layer	Surface of the external auditory canal, or into ducts of sebaceous glands	Cerumen, a waxy material	Soon after birth

Nails

Definition. Plates of tightly packed, hard, dead keratinized epidermal cells forming a clear, solid covering over the dorsal surfaces of the distal portions of the digits.

COMPONENTS OF A NAIL

Component	What it is
Nail body (plate)	The visible portion, comparable to the stratum corneum of the skin except the flattened keratinized cells fill with a harder keratin and are not shed. Mostly pink due to the vascularity of the tissue below
Free edge	The part extending beyond the finger or toe. White because there are no capillaries beneath it
Nail root	The portion buried in a fold of skin, not visible
Lunula	The little moon, a crescent-shaped area at the proximal end of the nail body. Vascular tissue does not show through because the epithelium there is thickened
Hyponychium	Thickened stratum corneum below the free edge. Secures the nail to the fingertip
Eponychium (cuticle)	A narrow band of epidermis extending from and adhering to the margin of the nail wall. It is stratum corneum
Nail matrix	Epithelium proximal to the nail root, containing dividing cells that produce new nail cells

Functions. Protect the ends of the digits, provide support and counter pressure to the palmar surface of the fingers which enhances touch perception and manipulation, and allow us to grasp and manipulate small objects, scratch, and groom.

Thick Skin and Thin Skin

The greatest contributor to epidermal thickness is an increased number of layers in the stratum corneum, which arises in response to greater mechanical stress.

STRUCTURAL DIFFERENCES

Feature	Thin (hairy) skin	Thick (hairless) skin
Distribution	All parts of the body except palms, palmar surface of digits, and soles	Palms, soles, palmar surface of digits
Epidermal strata	Stratum lucidum essentially lacking, thinner spinosum and corneum	Stratum lucidum present, thicker spinosum and corneum
Epidermal ridges	Lacking, because dermal papillae are poorly developed, fewer, and less organized	Present, because dermal papillae are well developed, more numerous, and organized in parallel rows
Hair follicles and arrector pili	Present	Absent
Sebaceous glands	Present	Absent
Sudoriferous glands	Fewer	More numerous
Sensory receptors	Sparser	Denser

Immune Protection, Sensation, Exchange

Two cell types with an immunological role

1. **Intraepidermal macrophages (Langerhans cells)**, alert the immune system to potentially harmful microbial invaders, recognizing and processing them.
2. **Macrophages in the dermis**, phagocytize bacteria and viruses that get past the intraepidermal macrophages of the epidermis.

Cutaneous sensations

Sensations that arise in the skin, including tactile sensations: touch, pressure, vibration, tickling, and thermal sensation. **Pain** is the sensation indicating impending or actual tissue damage. The receptors include tactile discs of the epidermis, corpuscles of touch, and hair root plexuses around each follicle.

Excretion and absorption

The skin has a small role in both. **Excretion** is the elimination of substances from the body. **Absorption** is the passage of materials from the external surface inward, and although the amount is negligible, it does occur.

What crosses. Lipid-soluble materials: the fat-soluble vitamins A, D, E, and K, certain drugs, gases such as oxygen and carbon dioxide, toxic materials, salts of heavy metals such as lead, mercury, and arsenic, and the oils of poison ivy and poison oak. Steroid drugs such as cortisone are lipid soluble and move into the papillary region of the dermis, where their anti-inflammatory effect includes inhibiting histamine production by mast cells. Other drugs are delivered through patches, nitroglycerin and fentanyl among them.

Synthesis of vitamin D

Only a small amount of exposure, 10 to 15 minutes twice a week, is necessary for vitamin D synthesis.

Clinical Correlations

Skin grafts

A **skin graft** is the transfer of a patch of healthy skin from a donor site to cover a wound. New skin cannot regenerate if the injury has damaged a large area of the stratum basale and its stem cells, which is exactly why the depth of a burn determines the treatment.

Benefits. Protects against fluid loss and infection, promotes tissue healing, reduces scar formation, prevents loss of function, and improves cosmetic outcome.

Complication. Tissue rejection. To minimize the risk, transplanted skin is taken from the same person, an **autograft**, or from an identical twin, an **isograft**. In severe burns where taking an autograft would cause more harm than good, **autologous skin transplantation** is used: small amounts of the person's epidermis are removed, keratinocytes are cultured in the lab and grown into thin sheets of skin, and those sheets are transplanted.

Albinism and vitiligo

Albinism is the inherited inability to produce melanin. Most affected individuals have melanocytes but cannot synthesize tyrosinase, so melanin is missing from skin, eyes, and hair. Manifestations include problems with night vision and a tendency for the skin to burn with sun overexposure.

Vitiligo is the partial or complete loss of melanocytes from patches of skin, producing irregular white spots.

Skin color as a diagnostic clue

FOUR COLOR CHANGES AND WHAT THEY SUGGEST

Finding	Appearance	What it suggests
Cyanosis	Blue	Blood is not picking up adequate oxygen from the lungs. Most apparent in mucous membranes, nail beds, and skin
Jaundice	Yellowish skin and sclera	A build-up of the yellow pigment bilirubin, likely indicating liver disease
Erythema	Redness	Engorgement of dermal capillaries with blood, from skin injury, heat exposure, infection, inflammation, or allergic reaction
Pallor	Paleness	May occur in conditions such as shock or anemia

These changes can be difficult to assess in individuals with darker skin. Examining the nail beds, gums, mucous membranes, and conjunctiva is the reliable approach.

Thermoregulation

THE SKIN'S TWO-DIRECTION RESPONSE

When the body is	Blood vessels	Sweat	Other
Too warm	Vasodilation brings warm blood to the surface so heat is lost to the environment	Sweat production increases, and evaporation cools the surface	Goal is to shed heat
Too cold	Vasoconstriction keeps blood away from the surface so heat is conserved in the core	Sweat production is reduced	Shivering generates heat through muscle activity

THE SKIN LAYERS MAP THE DEPTH OF A BURN

Burn classification is anatomy applied. A superficial burn involves the epidermis only, and it heals because the stratum basale is intact and its stem cells can repopulate the layers above. A partial-thickness burn reaches into the dermis, blisters because the dermal-epidermal junction fails, and still heals from surviving follicles and glands, since those structures are lined with epidermal cells that live deep in the dermis. A full-thickness burn destroys the dermis and its appendages, so there is no source of new epidermis left within the wound, and grafting becomes necessary. Read every burn question by asking one thing first: how deep did it go, and is the stratum basale still there.

Pre-work.

1 min

PART B

Four packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.

Pre-work · Intro to Anatomy

Intro to Anatomy. Read the Part A chapters first, then work through this in order. When you walk into class you should be ready to use this material, not hearing it for the first time.

Module 1 · 1 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the two intro chapters into mini visuals. Seven chunks, each in the shape that fits it.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1. **LADDER**

   ### The six levels of structural organization

   Six rungs, simplest at the bottom. Write one real example beside each rung, not the one from the notes.

2. **SPLIT**

   ### The seven directional pairs

   One split per pair. Pair on the branches, and underneath, the one sentence that proves you can use it. Star the two you are least sure of.

3. **GRID**

   ### Planes crossed with what they divide

   Planes down the side, the two parts each produces across the top. Six rows. Draw the cut line on a stick figure in one box.

4. **HUB**

   ### Body cavities

   Ventral and dorsal as two hubs. Spoke out to each subdivision, then to what it contains. Mark the diaphragm on the spoke it actually sits on.

5. **CHAIN**

   ### Serous membrane, wall to organ

   Body wall, arrow, parietal layer, arrow, cavity with fluid, arrow, visceral layer, arrow, organ. Then write the chain three times, once each for pericardium, pleura, peritoneum.

6. **HUB**

   ### The mediastinum

   Borders as spokes on one side, contents as spokes on the other. Five borders, five contents.

7 **GRID**

Quadrants and regions on one abdomen

Draw the abdomen once. Quadrant lines in one color, the nine-region lines in a second. Label the organs where they sit. One drawing does both systems.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

Anatomical position, with the planes on it

BRAIN DUMP FIRST

Two minutes. Anatomical position and every plane you can name, with what each one separates.

Draw a full figure in anatomical position. Mark the five features that define it: feet flat, arms at the sides, palms forward, head and eyes forward, body erect. Then cut the same figure with the sagittal, frontal, and transverse planes. Label each, and write the two portions each one creates.

In your notebook, head the page **M1-1 Anatomical position, with the planes on it** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can use midsagittal and parasagittal on one figure and say the difference in a sentence.

DRAWING 2

The cavities, lateral view

BRAIN DUMP FIRST

Two minutes. Every cavity, what is in it, and what separates it from the one next door.

Draw the body in profile. Outline the dorsal cavity and the ventral cavity, then subdivide each. Shade the thoracic and abdominopelvic cavities differently and draw the diaphragm between them. Put the mediastinum in, and shade the retroperitoneal space behind the peritoneum.

In your notebook, head the page **M1–2 The cavities, lateral view** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any organ on your drawing and say which cavity it sits in and which membrane wraps it.

DRAWING 3

Serous membrane, the fist and the balloon

BRAIN DUMP FIRST

One minute. Parietal, visceral, cavity, fluid. What each one touches.

Push a fist into a partly inflated balloon and draw that. The layer against your fist is visceral, the outer layer is parietal, the space between holds the fluid. Label all four. Then draw what happens to the space when it fills with air, and when it fills with fluid.

In your notebook, head the page **M1-3 Serous membrane, the fist and the balloon** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain a collapsed lung and a friction rub using only this drawing.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

The elbow is _______________ to the wrist.

Answer. Proximal.

Reasoning. Both structures are on the same limb, so the proximal and distal pair is available. Proximal means nearer the point of attachment to the trunk. The elbow attaches nearer the shoulder than the wrist does, so the elbow is proximal. Note the wrong answer you could reach for: the elbow is also superior to the wrist in anatomical position, but that is the weaker answer, because the question is about a limb and the limb-specific pair is the precise one.

Set A · Directional relationships

1. The sternum is _______________ to the vertebral column.
2. The lungs are _______________ to the diaphragm.
3. The thumb is _______________ to the little finger, in anatomical position.
4. The skin of the thigh is _______________ to the femur.
5. A stroke in the left cerebral hemisphere causes weakness on the right side of the body. That effect is _________________________.
6. Explain in one sentence why it is wrong to say the nose is proximal to the chin.

Set B · Planes and sections

1. A scan of the head shows both eyes and both ears in the same image, with the top of the skull missing. Which plane produced it, and how do you know?
2. A cut separates the body into anterior and posterior portions. Name the plane. _________________________
3. A section through the arm shows a single round bone surrounded by a ring of muscle. Which plane, and how do you know?
4. You want an image showing the relationship between the front of the heart and the sternum. Which plane serves you best, and why?

Set C · Cavities and membranes

1. A stab wound opens the pleural cavity. Predict what happens to the lung on that side, and say why.
2. A patient has a ruptured appendix. Name the membrane involved, the condition it produces, and why the problem does not stay local.
3. A surgeon plans to reach the right kidney. Is the approach through the peritoneal cavity or behind it? Name the term and one other structure in the same space.
4. A friction rub is heard over the heart. Name the two layers rubbing, the fluid that should be preventing it, and the condition.

Set D · Locate it

1. A patient reports pain in the right lower quadrant. Name two organs that could be responsible.
2. Give the regional term for the front of the elbow, the back of the knee, and the sole of the foot.
3. Sort as axial or appendicular: brachial, thoracic, popliteal, vertebral, carpal.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The six levels of structural organization

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

2

The seven directional pairs

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

3

Planes crossed with what they divide

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

Body cavities

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

5

Serous membrane, wall to organ

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

6

The mediastinum

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

7

Quadrants and regions on one abdomen

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

Pre-work · Cell Anatomy

Cell Anatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 1 · 2 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the cell chapter into mini visuals. Eight chunks, eight small visuals, each in the shape that fits it.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1. **HUB**

   ### The generalized cell

   One hub, three spokes: plasma membrane, nucleus, cytoplasm. Hang one job off each spoke. This is the frame everything else attaches to.

2. **GRID**

   ### Permeability

   Semipermeable, permeable, impermeable down the side. Definition and one example across the top. Three rows, and the examples must be yours.

3. **HUB**

   ### Inside the nucleus

   Nucleus at the center. Spoke to envelope, pores, chromatin, chromosomes, nucleoplasm, nucleolus. One job on each spoke.

4. **SPLIT**

   ### Cytoplasm versus cytosol

   One split. The trap is that one contains the other. Write the containing relationship on the branch, not just two definitions.

5. **CHAIN**

   ### The secretory pathway

   Nucleus, arrow, rough ER, arrow, vesicle, arrow, Golgi, arrow, vesicle, arrow, plasma membrane. Then write what happens to the protein at each arrow.

6 **GRID**

Rough ER against smooth ER

The two down the side. Structure, function, and what it sends onward across the top.

7 **HUB**

The cytoskeleton

Three spokes: microfilaments, intermediate filaments, microtubules. Protein on one line, job on the next. Circle the one that builds cilia.

8 **SPLIT**

9 plus 0 against 9 plus 2

Centrioles on one branch, cilia and flagella on the other. Write the arrangement and, underneath, the one word that separates them: movement.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The generalized cell, labelled

BRAIN DUMP FIRST

Two minutes. Every organelle you can name and one job for each.

Draw one cell large enough to label. Include the plasma membrane, nucleus with nucleolus, rough and smooth ER, Golgi, mitochondria, lysosomes, peroxisomes, ribosomes, cytoskeleton, centrioles. Label every one. Do not draw a diagram you have seen, draw the one you remember.

In your notebook, head the page **M1-4 The generalized cell, labelled** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any organelle on your drawing and name what fails if it stops working.

DRAWING 2

The plasma membrane, close up

BRAIN DUMP FIRST

One minute. Every component of the membrane and where it sits.

Draw a section of bilayer wide enough to show detail. Mark the hydrophilic heads and hydrophobic tails and which way each faces. Add an integral protein spanning the layer, a peripheral protein on the surface, cholesterol between the tails, and a glycoprotein with its sugar chain facing out. Label which side is extracellular.

In your notebook, head the page **M1–5 The plasma membrane, close up** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why a lipid-soluble drug crosses this and a charged ion does not, using only your drawing.

DRAWING 3

Trace the pathway on your cell

BRAIN DUMP FIRST

One minute. The route a secreted protein takes, in order.

Go back to your first drawing. In a second color, draw the route a secreted protein takes from the gene to the outside of the cell. Number each step. Mark where the protein is made, where it is modified, and how it moves between stations.

In your notebook, head the page **M1-6 Trace the pathway on your cell** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can tell the story out loud without looking, and your numbers are in the right order.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A drug destroys the enzymes inside lysosomes. Predict what accumulates and where.

Answer. Undigested macromolecules and worn organelles build up inside the cell.

Reasoning. Work from structure to consequence. The lysosome is the digestive organelle, so its job is breaking down macromolecules, damaged organelles, and pathogens. Remove the enzymes and nothing else takes over that job, so the material has nowhere to go and stays inside the cell. Cells with the highest turnover show it first. This is the mechanism behind Tay-Sachs, where one missing lysosomal enzyme lets lipid accumulate in nerve cells until the cell fails.

Set A · Structure to job

1. The organelle that assembles ribosomal subunits is the ______________________.
2. A cell that secretes large amounts of protein would be rich in ______________________.
3. A cell that detoxifies drugs would be rich in ______________________.
4. The organelle with its own DNA and ribosomes is the ______________________.
5. Name one cell type that is multinucleated and one that is anucleated.

Set B · Predict the failure

1. A mutation stops the cell producing glycoproteins. Predict two things the cell can no longer do well, and say why.
2. A cell cannot make enough ATP. Name the organelle involved, and name the three tissues that would show it first. Explain your choice.
3. The cilia lining the airway stop moving. Predict the clinical picture, and say which microtubule arrangement has failed.
4. Cholesterol is removed from the membrane. Predict what happens to the membrane as temperature changes.

Set C · Tell it apart

1. A student says cytoplasm and cytosol are the same thing. Correct them in one sentence.

2. Free ribosomes and bound ribosomes make different proteins. Say what decides which is used.

3. Chromatin and chromosomes are the same material. Say what changes between them and when.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The generalized cell

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

2

Permeability

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

3

Inside the nucleus

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

4

Cytoplasm versus cytosol

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

5

The secretory pathway

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

6

Rough ER against smooth ER

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

7

The cytoskeleton

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

8

9 plus 0 against 9 plus 2

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

Pre-work · Tissues and Histology

Tissues and Histology. Read the Part A chapter first, then work this in order. Most of this packet is pattern recognition, and pattern recognition is built by drawing, not by reading.

Module 1 · 3 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the histology chapter into mini visuals. Eight chunks, and the two grids do most of the work.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1. **HUB**

   ### The four tissue types

   One hub, four spokes. One job and one location per spoke. Everything else in this chapter hangs off this frame.

2. **GRID**

   ### Epithelium, layers crossed with shape

   Simple and stratified down the side. Squamous, cuboidal, columnar across the top. Fill each box with a location. Add pseudostratified and transitional as two extra rows and say why they do not fit the grid cleanly.

3. **GRID**

   ### The five cell junctions

   Junction down the side. Protein, job, and one location across the top. Five rows. Star the two you will confuse.

4. **SPLIT**

   ### Epithelium against connective tissue

   One split, three branches on each side: cells and matrix, blood supply, location. The contrast is the point.

5. **HUB**

   ### What connective tissue is made of

   Hub is extracellular matrix plus cells. Spoke to ground substance and to the three fibers, and separately to the six cell types. One job each.

6. **LADDER**

   ### Blasts and cytes

   Two rungs. Blasts on the lower rung building, cytes above maintaining. Write three examples of each pair beside the rungs.

7 CHAIN

How a gland releases its product

Three short chains, one per mode. Merocrine, apocrine, holocrine. Draw what happens to the cell at the end of each chain. That difference is the whole classification.

8 GRID

Cartilage and bone

Hyaline, elastic, fibrocartilage, compact bone down the side. Matrix, cells, perichondrium yes or no, and location across the top.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The eight epithelia, thumbnails

BRAIN DUMP FIRST

Two minutes. Every epithelium you can name, with its shape.

Divide the box into eight small frames. Sketch each epithelium as it looks on a slide: the basement membrane as a line, the cells above it in the right shape and number of layers. Add cilia, microvilli, goblet cells, and keratin where they belong. Label each and write one location under it. Small and fast, not beautiful.

In your notebook, head the page **M1-7 The eight epithelia, thumbnails** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

Someone shows you a slide and you can name it from the shape of the top layer and the number of layers, without reading a caption.

DRAWING 2

The five junctions on two cells

BRAIN DUMP FIRST

One minute. The five junctions and what each one attaches to.

Draw two adjacent cells sitting on a basement membrane. Put all five junctions on them in the right places: tight junction near the apex, adherens belt below it, desmosomes between the cells, hemidesmosomes at the basement membrane, gap junctions as channels. Label the protein at each.

In your notebook, head the page **M1–8 The five junctions on two cells** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain a blistering condition by pointing to which junction failed.

DRAWING 3

A connective tissue field

BRAIN DUMP FIRST

One minute. Every fiber and every cell type you can name.

Draw a field of loose areolar tissue. Show collagen fibers as thick wavy bundles, elastic fibers as thin branching threads, and ground substance as the space between. Add a fibroblast, a macrophage, a mast cell, an adipocyte, and a plasma cell, and label each. Then in a second color, redraw the same field as dense regular tissue and say what changed.

In your notebook, head the page **M1-9 A connective tissue field** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can look at your two versions and say why one heals fast and the other heals slowly.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A slide shows several layers of cells, and the cells at the free surface are flat. What is it, and where would you expect to find it?

Answer. Stratified squamous epithelium. Skin if keratinized, otherwise esophagus, mouth, or vagina.

Reasoning. Name the layers first, then the shape at the free surface. Several layers means stratified. The cells at the top are flat, so squamous. That is the full name, and the naming rule always reads the apical layer, never the basal one. Then ask what the tissue is for: many layers that shed from the top means it survives abrasion, so look for it wherever something rubs.

Set A · Name it from the description

1. A single layer of tall cells with goblet cells and microvilli. Name it and give one location.
2. Several layers where all cells touch the basement membrane but the nuclei sit at different heights. Name it.
3. Cells that change shape from dome to flat as the organ fills. Name it and the organ.
4. A single layer of flat cells one cell thick. Name it and say what that thinness makes possible.

Set B · Junctions and consequences

1. A condition destroys desmosomes in the epidermis. Predict what happens to the skin and say why.
2. A toxin opens the tight junctions in the intestinal lining. Predict what can now cross that could not before.
3. Cardiac muscle relies on two junction types at the intercalated disc. Name both and say what each contributes.

Set C · Connective tissue and glands

1. A tendon and a skin wound are injured on the same day. Predict which heals faster and explain using vascularity.

2. A gland destroys its own cells to release its product. Name the mode and one gland that uses it.

3. A patient has an allergic reaction with swelling and redness. Name the connective tissue cell responsible and the substance it released.

4. Sort by matrix, softest to hardest: bone, blood, cartilage, areolar tissue. Say what changes across that order.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The four tissue types

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

2

Epithelium, layers crossed with shape

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

3

The five cell junctions

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

Epithelium against connective tissue

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

5

What connective tissue is made of

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

6

Blasts and cytes

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

7

How a gland releases its product

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

8

Cartilage and bone

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

Pre-work · Integumentary System

Integumentary System. Read the Part A chapter first, then work this in order. This is the first organ where all four tissue types show up together, so it is where the module gets used rather than listed.

Module 1 · 4 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the integumentary chapter into mini visuals. Seven chunks, and the ladder is the one you will use on every exam.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1. **LADDER**

The five strata

Five rungs, basale at the bottom. Beside each rung write what the keratinocyte is doing at that level. The mnemonic goes in the margin, not on the ladder.

2. **HUB**

The four cells of the epidermis

One hub, four spokes: keratinocyte, melanocyte, intraepidermal macrophage, Merkel cell. Percentage on one line, job on the next. Mark which stratum each lives in.

3. **CHAIN**

Melanin transfer

Melanocyte, arrow, projection between keratinocytes, arrow, granules enter the keratinocyte, arrow, granules cluster over the nucleus. Draw the last step, do not just write it. That veil is the whole point.

4. **SPLIT**

Papillary against reticular dermis

One split. Fraction of the dermis, fibers, and what lives in each on the branches. Add the three structures inside a dermal papilla.

5. **GRID**

The four skin glands

Sebaceous, eccrine, apocrine, ceruminous down the side. Where they are, what they secrete, where the duct opens, and when they switch on across the top.

6 **HUB**

The nail

Nail at the center. Spoke to body, free edge, root, lunula, hyponychium, eponychium, matrix. One line each on what it is or does.

7 **CHAIN**

Thermoregulation, both directions

Two chains from one starting point. Too warm goes one way, too cold the other. Put vessels and sweat on both chains and shivering on one.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The skin in cross section

BRAIN DUMP FIRST

Two minutes. Every layer and every structure in the skin.

Draw the full thickness: epidermis, dermis with papillary and reticular regions, and hypodermis. Add a hair follicle running down, a sebaceous gland on it, an arrector pili muscle, an eccrine gland with its duct reaching the surface, dermal papillae with capillary loops and Meissner corpuscles, and a lamellated corpuscle in the hypodermis. Label everything.

In your notebook, head the page **M1–10 The skin in cross section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can draw a horizontal line across your own drawing at any depth and say what a burn to that depth would destroy.

DRAWING 2

The five strata, cell by cell

BRAIN DUMP FIRST

One minute. The strata in order and what changes between them.

Zoom in on the epidermis alone. Draw the five strata stacked, with the cells changing shape as they rise: cuboidal and dividing at the base, spiny in the middle, granular above, clear, then flat dead squames at the top. Mark where the nucleus disappears and where lamellar granules release their sealant. Mark which layer is missing in thin skin.

In your notebook, head the page **M1–11 The five strata, cell by cell** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why the epidermis survives without a blood supply, using only your drawing.

DRAWING 3

Hair follicle and bulb

BRAIN DUMP FIRST

One minute. The parts of a hair and everything attached to the follicle.

Draw one follicle from the surface down to the bulb. Show the shaft above the skin and the root below, the three concentric layers, the external and internal root sheaths, the dermal root sheath, and at the base the bulb with the papilla and the matrix. Attach the sebaceous gland, the arrector pili, and the hair root plexus.

In your notebook, head the page **M1–12 Hair follicle and bulb** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say where the hair actually grows from and why cutting it does not change that.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A burn destroys the epidermis and the upper dermis but leaves hair follicles and glands intact. Predict whether it regenerates, and say why.

Answer. Yes, it regenerates, from the follicles and glands.

Reasoning. Ask one question of every burn: is there a source of new epidermal cells left inside the wound. The stratum basale is gone at the surface, but hair follicles and glands are lined with epidermal cells and they sit deep in the dermis, so they survive. New epithelium spreads outward from each surviving structure until the sheets meet. A full-thickness burn destroys those too, which is why it cannot heal from the middle and needs grafting.

Set A · Layers and cells

1. Name the strata in order, deep to superficial, for thick skin.

   ____

2. Which stratum is missing in thin skin, and where on the body would you find thick skin?

   ____

3. The cell that alerts the immune system to an invader in the epidermis is the ____________________.
4. Which epidermal cell is least numerous, where does it sit, and what does it contact?

   ____

Set B · Color and pigment

1. Two people have very different skin color but the same number of melanocytes. Explain the difference.

   ____

   ____

   ____

2. A patient appears yellow in the skin and the sclera. Name the finding, the pigment, and the organ you would suspect.

   ____

   ____

   ____

3. A patient with dark skin is suspected of being cyanotic. Say where you would look instead, and why.

   ____

   ____

   ____

4. Explain why a tan is protective and why melanocytes are still damaged by the same exposure.

   ____

   ____

   ____

Set C · Structures and consequences

1. A gland opens onto a hair follicle in the axilla and switches on at puberty. Name it. ________________________
2. Damage to the nail matrix. Predict the effect on the nail and say why.
3. A patient loses the ability to sweat over a large area. Predict the consequence in hot weather and name the function lost.
4. Explain why a steroid cream works on inflamed skin but a large protein would not cross at all.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The five strata

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

2

The four cells of the epidermis

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

3

Melanin transfer

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

4

Papillary against reticular dermis

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

5

The four skin glands

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

6

The nail

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

7

Thermoregulation, both directions

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

Lab Structure List.

1 min

PART C

Part C of the Module 1 packet. These are the lab structures and concepts to learn for the lab portion of Exam 1. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

Introduction to Anatomy

3 min

Body landmarks, orientation, planes, cavities, membranes, and the microscope. Everything here is testable on models, images, the cadaver, and slides.

1. **All anterior and posterior body regional landmarks.** Know the term, and be able to point to it on a model and on yourself.

Anterior regions

ANTERIOR REGIONAL TERMS

Term	Region	Term	Region
Frontal	Forehead	**Inguinal**	Groin
Cranial	Skull or head	**Pubic**	Area near the genital region
Facial	Face	**Brachial**	Upper arm
Orbital or ocular	Eye region	**Antebrachial**	Forearm
Buccal	Cheek	**Antecubital**	Front of the elbow
Otic	Ear region	**Carpal**	Wrist
Nasal	Nose	**Pollex**	Thumb
Oral	Mouth region	**Palmar**	Palm of the hand
Mental	Chin	**Digital or phalangeal**	Fingers or toes
Cervical	Neck region	**Patellar**	Kneecap
Thoracic	Chest area	**Crural**	Leg, the shin area
Sternal	Breastbone	**Tarsal**	Ankle
Mammary	Breast region	**Pedal**	Foot
Abdominal	Below the chest, above the pelvis	**Hallux**	Big toe
Umbilical	Navel or belly button	**Axillary**	Armpit
Pelvic	Lower abdomen or hip area	**Coxal**	Hip
Femoral	Thigh		

Posterior regions

POSTERIOR REGIONAL TERMS

Term	Region	Term	Region
Cephalic	Head	**Sacral**	Area between the hips, over the sacrum
Occipital	Back of the head	**Gluteal**	Buttock region
Cervical	Neck	**Femoral**	Thigh
Acromial	Point of the shoulder	**Popliteal**	Back of the knee
Scapular	Shoulder blade	**Sural**	Calf
Dorsal	Back	**Fibular or peroneal**	Lateral side of the leg
Vertebral	Spinal column	**Calcaneal**	Heel
Olecranal	Back of the elbow	**Plantar**	Sole of the foot
Lumbar	Lower back or loin area		

2. **Identify those structures in terms of body orientation and direction.** Be able to use each in a sentence. The trachea is ventral to the spine. The nose is medial to the cheekbone.

DIRECTIONAL TERMS, IN PAIRS

Term	Its opposite	What the pair describes
Anterior (ventral)	Posterior (dorsal)	Toward the front, versus toward the back
Superior	Inferior	Above, versus below
Medial	Lateral	Toward the midline, versus away from it
Cephalad (cranial)	Caudal	Toward the head, versus toward the tail end
Proximal	Distal	Nearer the trunk, versus farther from it. Limbs only
Superficial	Deep	Toward the body surface, versus away from it
Ipsilateral	Contralateral	On the same side of the body, versus the opposite side

3. **Body planes and sections.** Midline, median (midsagittal), parasagittal, coronal (frontal), transverse (horizontal or cross-sectional), oblique.

4. **Body cavities.** Ventral and dorsal, superior and inferior, cranial, vertebral, thoracic, abdominopelvic, mediastinum.

   What is the name of the structure that separates the thoracic cavity from the abdominopelvic cavity?

   ____

5. **Mediastinum.**

   1. Know the structures that form the following borders: superior, inferior, ventral, dorsal, and lateral.

      ____

      ____

   2. Know the contents of the mediastinum.

      ____

6. **Retroperitoneum.** Name the retroperitoneal structures. Memory device: SADPUCKER.

7. **Serous membranes and cavities.** Pericardial (pericardium): parietal, visceral, cavity, fluid. Peritoneal (peritoneum): parietal, visceral, cavity, fluid. Pleural (pleura): parietal, visceral, cavity, fluid.

8. **Abdominopelvic quadrants and the major structures found within each.**

THE FOUR QUADRANTS AND THEIR ORGANS

Quadrant	Organs found within
Right upper (RUQ)	Liver (right lobe), gallbladder, right kidney, duodenum, head of the pancreas, hepatic flexure of the colon, part of the ascending and transverse colon
Left upper (LUQ)	Stomach, spleen, left lobe of the liver, left kidney, body and tail of the pancreas, splenic flexure of the colon, part of the transverse and descending colon
Right lower (RLQ)	Cecum, appendix, ascending colon, right ureter, right ovary and uterine tube, right spermatic cord
Left lower (LLQ)	Descending colon, sigmoid colon, left ureter, left ovary and uterine tube, left spermatic cord

The urinary bladder and the uterus sit at the midline in the hypogastric region and can be described in either lower quadrant.

9. **Be able to calculate total magnification, and name the parts of the microscope.**

CHAPTER 2

Epithelial Tissues

1 min

Identify the tissue on the slide, then write its function and location. Tick each structure as you find it.

General structures, on every epithelial slide

STRUCTURES TO IDENTIFY

- ☐ Apical surface
- ☐ Lateral surface
- ☐ Basal surface
- ☐ Basement membrane
- ☐ Lamina propria, when present
- ☐ Nuclei
- ☐ Lumen or free surface
- ☐ Cilia, when present
- ☐ Microvilli, when present
- ☐ Goblet cells, when present
- ☐ Keratin, when present

☐ Simple squamous

FUNCTION ______________________ LOCATION ______________________

☐ Keratinized stratified squamous

FUNCTION ______________________ LOCATION ______________________

☐ **Non-keratinized stratified squamous**

FUNCTION ______________________ LOCATION ______________________

☐ **Simple cuboidal**

FUNCTION ______________________ LOCATION ______________________

☐ **Stratified cuboidal**

FUNCTION ______________________ LOCATION ______________________

☐ **Simple columnar**

FUNCTION ______________________ LOCATION ______________________

☐ **Pseudostratified ciliated columnar**

FUNCTION ______________________ LOCATION ______________________

☐ **Transitional**

FUNCTION ______________________ LOCATION ______________________

Which tissues typically contain cilia? ______________________ Function: ______________________

Microvilli? ______________________ Function: ______________________

Goblet cells? ______________________ Function: ______________________

CHAPTER 3

Connective Tissues

1 min

Connective tissue proper first, then cartilage and bone. For Module 1 you need the basic identification. The structures inside cartilage and bone come back in Module 2.

Connective tissue proper

☐ **Loose areolar**

FUNCTION ______________________ LOCATION ______________________

STRUCTURES TO IDENTIFY

- ☐ Collagen fibers
- ☐ Fibroblasts
- ☐ Elastic fibers
- ☐ Mast cells

☐ **Adipose**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Adipocytes ☐ Fat vacuoles containing lipids ☐ Blood vessels

☐ **Reticular**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Reticular fibers

☐ **Dense regular**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Collagen fibers ☐ Fibroblasts

☐ **Dense irregular**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Collagen fibers ☐ Fibroblasts

☐ **Elastic**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Elastin fibers

Cartilage and osseous tissue BASIC ID IN MODULE 1 STRUCTURES IN MODULE 2

Name the tissue, say what it does and where it is. Tick the structures you can already find, the rest is Module 2 work.

☐ **Hyaline cartilage**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Perichondrium ☐ Chondrocytes
☐ Lacunae ☐ Extracellular matrix

☐ **Elastic cartilage**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

☐ Perichondrium ☐ Chondrocytes ☐ Elastin fibers
☐ Lacunae ☐ Extracellular matrix

- [ ] **Fibrocartilage**

FUNCTION ______________________ LOCATION ______________________

STRUCTURES TO IDENTIFY

- [ ] Lacunae
- [ ] Extracellular matrix
- [ ] No perichondrium
- [ ] Chondrocytes
- [ ] Collagen fibers

- [ ] **Osseous tissue**

FUNCTION ______________________ LOCATION ______________________

STRUCTURES TO IDENTIFY

- [ ] Osteons
- [ ] Lacunae
- [ ] Osteocytes
- [ ] Central (Haversian) canal
- [ ] Lamellae
- [ ] Canaliculi

CHAPTER 4

Muscle Tissues

1 min

Basic identification for Module 1. Name the tissue, what it does, where it is, and whether it is voluntary or involuntary. The sarcomere and the connective tissue wrappings come back in Module 3.

- [ ] **Skeletal muscle**

FUNCTION ______________________ LOCATION ______________________

VOLUNTARY OR INVOLUNTARY ______________________

STRUCTURES TO IDENTIFY

- [ ] Striations, dark and light
- [ ] Nuclei, multiple and peripheral
- [ ] Sarcolemma

- [ ] **Cardiac muscle**

FUNCTION ______________________ LOCATION ______________________

VOLUNTARY OR INVOLUNTARY ______________________

STRUCTURES TO IDENTIFY

- [ ] Branching fibers
- [ ] Single central nucleus
- [ ] Intercalated discs
- [ ] Striations

- [ ] **Smooth muscle**

FUNCTION ______________________ LOCATION ______________________

VOLUNTARY OR INVOLUNTARY ______________________

STRUCTURES TO IDENTIFY

- [ ] Spindle-shaped cells
- [ ] Single central nucleus
- [ ] No striations

What two cell junctions are found at intercalated discs, and what does each one do?

____

____

CHAPTER 5

Nerve Tissues

1 min

Basic identification for Module 1. The detail comes back in Module 5.

- [ ] **Multipolar neuron**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Cell body (soma)
- [ ] Nucleus
- [ ] Nucleolus
- [ ] Dendrites
- [ ] Axon
- [ ] Axon hillock
- [ ] Synaptic bulb

- [ ] **Spinal cord**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] White matter
- [ ] Grey matter
- [ ] Ventral horn
- [ ] Dorsal horn
- [ ] Lateral horn, when present

- [ ] **Cerebral cortex**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Pyramidal cells

- [ ] **Cerebellum**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Purkinje cells

- [ ] **Nerve, cross section**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Axon
- [ ] Schwann cells
- [ ] Nodes of Ranvier
- [ ] Epineurium
- [ ] Perineurium
- [ ] Endoneurium

CHAPTER 6

Blood Vessels and Formed Elements

1 min

Basic identification for Module 1. The detail comes back in Module 3.

Vessel walls

- [ ] **Artery**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Tunica intima
- [ ] Tunica media
- [ ] Tunica externa
- [ ] Endothelium
- [ ] Smooth muscle
- [ ] Lumen
- [ ] Internal elastic lamina
- [ ] Dense elastic connective tissue
- [ ] Erythrocytes

- [ ] **Vein**

FUNCTION ____________________ LOCATION ____________________

STRUCTURES TO IDENTIFY

- [ ] Tunica intima
- [ ] Tunica media with smooth muscle
- [ ] Lumen
- [ ] Endothelium
- [ ] Tunica externa
- [ ] Erythrocytes

Peripheral blood smear

Identify each formed element, then write its function and how you recognized it.

- [ ] **Neutrophil**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Basophil**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Eosinophil**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Monocyte**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Lymphocyte**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Erythrocyte**

FUNCTION ____________________ LOCATION ____________________

- [ ] **Thrombocyte**

FUNCTION ____________________ LOCATION ____________________

CHAPTER 7

Integumentary System

1 min

Layers in order, deep to superficial, then the structures in the dermis and hypodermis.

Strata of the epidermis

☐ **Stratum corneum**

FUNCTION ______________________ LOCATION ______________________

☐ **Stratum lucidum**

FUNCTION ______________________ LOCATION ______________________

☐ **Stratum granulosum**

FUNCTION ______________________ LOCATION ______________________

☐ **Stratum spinosum**

FUNCTION ______________________ LOCATION ______________________

☐ **Stratum basale**

FUNCTION ______________________ LOCATION ______________________

Dermis

☐ **Dermis, on slides and models**

STRUCTURES TO IDENTIFY

- ☐ Papillary layer
- ☐ Reticular layer
- ☐ Areolar connective tissue
- ☐ Dense irregular connective tissue
- ☐ Dermal papillae
- ☐ Eccrine sweat gland and duct
- ☐ Hair
- ☐ Hair follicle
- ☐ Hair root
- ☐ Hair bulb
- ☐ Hair shaft
- ☐ Arrector pili muscle
- ☐ Sebaceous gland and duct
- ☐ Apocrine gland and duct
- ☐ Meissner's corpuscles
- ☐ Free nerve endings
- ☐ Capillary loops

Hypodermis, the subcutaneous tissue

☐ **Hypodermis**

STRUCTURES TO IDENTIFY

- ☐ Adipose connective tissue
- ☐ Areolar connective tissue
- ☐ Lamellated (Pacinian) corpuscle

Name the three structures found within dermal papillae.

____

Which epidermal layer is missing in thin skin? ______________________